

CHEN LI

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science

Chicago, IL

Expected Dec 2026

Relevant Coursework: Distributed Systems, Databases, Data Mining, Deep Learning, Machine Learning, Computer Graphics

University of Wisconsin–Madison

Bachelor of Science in Computer Science

Madison, WI

May 2025

Award: 1st Place Worldwide — RoboCup 2024 SPL Challenge Shield

EXPERIENCE

ByteDance — (Enterprise AI) Software Engineer Intern

San Jose, CA | Jun 2026 – Present

- Designed and shipped a declarative access-control reconciler that automatically synchronizes permission policies across a distributed data platform, replacing manual provisioning for multiple production pipelines and validating security boundaries through systematic test scenarios.
- Built at-least-once streaming pipelines on Apache Flink processing millions of events per day, with end-to-end correctness guarantees powering downstream analytics.
- Delivered API integrations and refactored query pipelines for a large-scale analytics system, reducing recurring production incidents by ~30%.
- Built a Go (Cobra) CLI with layered command groups and version-locked extensions for internal AI tooling, enforcing command validity and version parity through CI gates.

DataLynn — Software Engineer Intern

Remote | Jun 2025 – Aug 2025

- Designed a modular FastAPI LLM backend for a natural-language-to-sql platform, routing sql queries to the correct module.
- Fine-tuned a T5 model with PEFT and a custom tokenizer to 80% accuracy on query-intent classification.
- Built automated data-generation pipelines (LangChain + OpenAI API) producing 5,000+ synthetic JSON samples to accelerate training.

BaoSight — Software Engineer Intern

Shanghai, China | May 2024 – Aug 2024

- Engineered a reinforcement-learning digital twin system in Python modeling real-time steel-manufacturing logistics for intelligent crane scheduling.
- Trained a PPO agent to a zero-collision rate in simulation, validated against industry benchmarks with senior researchers.

University of Wisconsin–Madison (Prof. Wei) — Research Assistant

Madison, WI | Jun 2023 – May 2025

- Built an end-to-end ML pipeline for Raman-spectrum data (characteristic-peak extraction, noise reduction) for PAH analysis.
- Trained CNN, DNN, Random Forest, and SVR models predicting PAH composition at 93%+ accuracy; built a custom NumPy/SciPy classifier reaching 100% non-PAH identification.

UW-Madison RoboCup Lab, UW–Madison — Research Assistant

Madison, WI | Jan 2023 – Apr 2025

- Designed and trained a centralized multi-agent PPO policy enabling robots to pass, dribble, and kick, integrated into a C++ codebase running on physical robots.
- Tuned tactics and coordination on-site at the RoboCup 2024 world finals, contributing to a 1st-place-worldwide finish.
- Co-authored the resulting methodology paper accepted to ICRA 2025.

PUBLICATIONS

- Labiosa, A., et al. "Reinforcement Learning Within the Classical Robotics Stack: A Case Study in Robot Soccer" (ICRA 2025)
- Wu, Z., et al. "A Two-Step Artificial Intelligence Framework for Discriminating and Quantifying Mixed Pollutants" (ACS EST Water 2025) (co-1st author)

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL (Postgres), JavaScript, R, HTML/CSS

ML / AI: PyTorch, TensorFlow, PEFT, LangChain, OpenAI API, RLHF, PPO, OpenCV, scikit-learn

Data & Distributed: Flink, Spark, Hadoop, Kafka, Airflow, ByteHouse/ClickHouse, Paimon, pandas, NumPy

Backend & Cloud: FastAPI, Node.js, REST APIs, RBAC, KMS, Docker, Terraform, AWS, Azure, Git